



Jaehyun Byun

Kyung Hee University
B.S. in Computer Science

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EDUCATION

- Kyung Hee University, Seoul2020.03 - 2026.02
- B.S. in Computer ScienceGPA: 3.8/4.5

TECHNICAL SKILLS

Languages: C++, Python, C#, SQL
Developer Tools: Jira, Git, Unity, Unreal, Arduino, TensorFlow, Hugging Face
Design Tools: Figma, Blender, Miro
Coursework: HCI, UI&UX Programming, Computer Graphics, Algorithms, Multimodal Data Processing
Areas of Interest: Human-Computer Interaction, Extended Reality, Multimodal AI
Certifications: SQL Developer (Korea Data Agency), AI Prompt Engineering Professional (Korea Productivity Center)

AWARDS & HONORS

- ACM CHI Student Design Competition Finalist (Top 4)2025.05
- First Author and Team Leader, ACM CHI Conference on Human Factors in Computing Systems
- Ministry of Science and ICT 2024 XR Device Contents Makerthon Excellence Award2024.12
- President’s Award, National IT Industry Promotion Agency (NIPA)

PUBLICATIONS

- CrossGaussian: Enhancing Remote Collaboration through 3D Gaussian Splatting and Real-time 360° Streaming - Jaehyun Byun, Byunghoon Kang, Yonghyun Gwon, Hongsong Choi, Yunseo Do, Eunho Kim, Sangkeun Park, and Seungjae Oh *UIST Adjunct 2025* (ACM Symposium on User Interface Software and Technology)
- Le Petit Care: A Child-Attuned Design for Personalized ADHD Symptom Management Through AI-powered Extended Reality - Jaehyun Byun, Chowon Joung, Yerim Lee, Suyun Lee, and Woogy Won *CHI EA 2025* (Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems)
- Metaverse Platform Specialized in Discussion and Debate Functions in Online Communication Systems - Minsu Kim, Jaehyun Byun, and Tack Woo *Journal of Digital Contents Society*, Vol. 25, No. 6, Jun. 2024

PROFESSIONAL SERVICES

- Reviewer, IEEE Conference on Virtual Reality and 3D User Interfaces2026
- Honored Partner, Vietnam Mobile Summit2024

VOLUNTEER

- Online Education Mentor, KT Group2025.05 - 2025.11
- Provided online remote mentoring in CS and programming for educationally underserved students

PROJECTS

- CrossGaussian, ITEM Lab (Kyung Hee University)2024.09 - 2025.10
- Real-Time Remote Collaboration Interaction Research Based on Camera Based Room-Scale Neural Rendering
- Tools & Technologies: Azure DevOps, C++, C#, Python, HLSL, Unity, Photon, CUDA, FFmpeg
- Role & Team: Research Lead | Client ×2, Backend ×2, Data Analysis ×1
- Details: Addressed information asymmetry in remote collaboration by integrating room-scale 3DGS reconstruction with 360° video streaming; designed visualization and interaction design space through prior literature review; automated data collection and remote reconstruction pipeline by bridging Unity/HLSL client with Python/CUDA backend via SSH/SFTP; implemented rendering-based interaction techniques and validated hypotheses through N=24 user study with statistical analysis; accepted as poster at ACM UIST 2025 and Demo at IEEE ISMAR 2025
- i-There, P&C Solution2024.06 - 2025.05
- AR fairy-tale app with AI-driven real-time ADHD symptom detection and personalized behavioral intervention
- Tools: Jira, C#, Python, Unity, MariaDB, TensorFlow, Blender, Figma
- Role & Team: PM & Lead Developer | Planning ×2, Design ×3, Programming ×2, Medical AI ×1
- Details: Structured 18 DSM-5 diagnostic criteria into digital behavioral metrics and designed user flow and art direction targeting children aged 8–12; implemented child-facing play content, built medical CNN model and deployed on-device via ONNX conversion; preprocessed and normalized user behavior and play data for AI consumption and built centralized DB-based log collection and processing pipeline; developed on P&C Solution’s industrial-grade METALENSE2 AR device and won Excellence Award at MSIT XR Device Contents Makeathon 2024